Project 3 - DSA 8020

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1. The following steps are needed to extract the annual mean temperatures at the specified location from the original dataset with daily temperatures at thousands of locations: . 1

Project Description

The ERA-Interim is a global atmospheric reanalysis dataset. Reanalysis refers to the process of producing spatially and temporally gridded datasets through data assimilation, primarily for climate monitoring and analysis. In the R data file **NA_MaxT_ERA.RData**, you will find daily maximum temperature values from *1979* to *2017*, with a spatial resolution of approximately 80 km. The original global dataset has been subsetted to a region covering the contiguous United States and extending into Canada and Mexico.

In Problems 1 and 2, you are required to analyze an ERA-Interim average temperature time series at yearly and monthly temporal resolutions at a spatial location of your choice.

Problem 1. Analyzing Yearly Average Temperatures Time-Series

1. The following steps are needed to extract the annual mean temperatures at the specified location from the original dataset with daily temperatures at thousands of locations:

```r
load("NA_MaxT_ERA.RData")
NA_MaxT <- NA_MaxT - 273.15 # change from Kelvin to Celsius
nlon <- length(NA_lon); nlat <- length(NA_lat)
nmon <- 12; nyr <- 39 # numbers of month/year
```

a. First, use the following R script to load the R data object:

```r
# Clemson Example
library(fields)
```

b. Second, choose a location of interest (with the information of longitude and latitude) and identify the nearest ERA grid cell to the chosen location. Below you can find a script to get the correct ERA grid cell for Clemson:

```
## Loading required package: spam

## Spam version 2.11-1 (2025-01-20) is loaded.
## Type 'help( Spam)' or 'demo( spam)' for a short introduction
## and overview of this package.
## Help for individual functions is also obtained by adding the
## suffix '.spam' to the function name, e.g. 'help( chol.spam)'.

##
## Attaching package: 'spam'

## The following objects are masked from 'package:base':
##
##     backsolve, forwardsolve

## Loading required package: viridisLite

##
## Try help(fields) to get started.
```

```r
Clemson.lon.lat <- c(-82.8374, 34.6834)
# You can get lon/lat infromation from google
dist2Clemson <- rdist.earth(
  matrix(Clemson.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
  )
id <- which.min(dist2Clemson)
(lon_id <- id %% nlon)
```

```
## [1] 58
```

```r
(lat_id <- id %/% nlon + 1)
```

```
## [1] 15
```

```r
# Check
(rdist.earth(matrix(Clemson.lon.lat, 1, 2),
matrix(c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2), miles = F)
== min(dist2Clemson))
```

```
##      [,1]
## [1,] TRUE
```

```r
library(fields)
Monterey.lon.lat <- c(-121.8979, 36.5973)
dist2Monterey <- rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
  )

id <- which.min(dist2Monterey)

(lon_id <- id %% nlon)
```

```
## [1] 5
```

```r
(lat_id <- id %/% nlon + 1)
```

```
## [1] 18
```

```r
(rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  matrix( c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2),
  miles = F
  )== min(dist2Monterey))
```

```
##      [,1]
## [1,] TRUE
```